

Amritraj Lamba

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EDUCATION

- AUGUST 2024-MAY 2026** **IB Diploma (Grade 11 and Grade 12)**
Heritage International Xperiental Learning, Gurugram.
Relevant Coursework: HL Math Analysis and Approaches, HL Chemistry, HL Physics, SL Economics, SL French B, SL English A Language and Literature
- Jul 2023 - Jun 2024** **IGCSE (Grade 10)**
10/10 A*, School Topper in Additional Mathematics, School Topper in Chemistry
- Standardized Test Scores**
SAT- 1550 (Maths 800, Reading&Writing 750)
AP- Calculus BC: 5/5
IELTS - Band 8

LEADERSHIP

- Founder, Sewage Suraksha (April 2024- Present)**
Developed a sewage lid with multi-gas sensors; it is a low-cost retrofit ensuring safety of municipal workers.
The lid detects harmful gases, including H₂S, CH₄, NH₃, & CO, and alerts authorities to hazardous gas levels. It has been tested on 17 sewers and has been cleared by the National Disaster Management Authority for adoption.
- Founder, Flood Shield (July 2024-Present)**
Flood detection IOT for real-time cloud mapping and emergency coordination; Special Praise for "practicality" at the **Princetonian**. Adopted by the National Disaster Management Authority.
Designed and piloted an affordable early warning system for floods that alerts authorities to areas where flooding occurs; adopted by 1000+ houses. Commended by the Haryana Government.
Also received the **Crest Gold Award** for this invention

**Founder and Leader—Ingenix School STEM Society
(October 2024- October 2025)**

Revived and expanded society from 20 to 50+ members.

Mentored 2 teams for the Cambridge Science Competition & F1 in Schools, both qualified internationals.

**Leader - “FUNCTION”, After School Robotics Club
October 2024-October 2025**

Founded a 20+ member club teaching CAD, programming, and mechanical design outside of competition teams

Introduced collaborative engineering workshops (bridge/tower builds, quizzes) to strengthen teamwork and experimentation

Built a hands-on learning culture that became an entry pathway for beginners; several members progressed to competitive robotics

RESEARCH

Automation & Robot Adoption in Delhi-NCR Factories,

Author

January 2025- September 2025

Analysed five factories from fully manual (e.g., Metaval Engineering) to highly automated (e.g., Kadence Automation) to assess efficiency, training needs, and scalability

Found robot-assisted systems improved efficiency by up to 20% while reducing fatigue and errors; highlighted the need for retraining, ergonomic design, and phased adoption

Developed a roadmap for SME modernisation without worker displacement; validated through Delhi-NCR field data, mentorship from Stanford’s Dr. Arvind Karunakaran, and **publication in Lex Localis**

Coral Reefs as Natural Barriers to Coastal Erosion in India,

Author March 2024- August 2025

Used satellite imagery and GIS to show shorelines protected by healthy reefs had less than half the erosion of areas with degraded reefs.

Provided evidence to integrate reef restoration into climate adaptation strategies for cities like Mumbai and Chennai, complementing engineered defenses.

Recognition included **publication in the International Journal of Environmental Sciences** and mentorship from Prof. Umang Soni, with validation in competitive research platforms.

SUMMER PROGRAMS

**University of Pennsylvania, Wharton Global Youth Program
June 2025**

Product Design Academy

Collaborated with Professor Taylor Caputo and international peers under the Wharton faculty, learning design thinking, prototyping and TAM/SAM/SOM analysis.

Learned to apply structured market-sizing frameworks to test creative engineering ideas against commercial viability.

Prototypes developed during the program reflected this rigorous approach, and faculty feedback validated the professional grounding in product innovation.

LaunchX, Online Bootcamp (June 2025)

Selected for LaunchX; collaborated with peers to ideate, prototype, and pitch ventures, gaining hands-on experience in market validation, business modelling, and product design.

INTERNSHIPS

Summer Intern, Vardan Envirolabs, Gurugram.

December 2024- February 2025

Identified available and capable employees as part of the talent search module team, designed to assist talent recruitment in companies.

Conducted market research for the firm, tapping into new domains such as the legal sector, to streamline services and the employment data collection process.

ADVANCED COURSES

Euler's Circle (June- July 2024):

"Serious Mathematics for Serious High School Students " under Stanford Prof. Simon Salzedo.

One of 30 members worldwide.

Capstone on linear algebra and projective geometry; improved modeling intuition later applied to CFD and robot mechanisms. Attended on 50% scholarship

Art of Problem Solving (AoPS)

Completed All Introductory & Intermediate Courses on Algebra, Number Theory, Geometry and Counting & Probability

ACADEMIC HONOURS AND AWARDS

MATHEMATICS

USA Math Olympiads (2022-24)

AMC 12B (2024): Scored 132/150

School Rank 1, Country Rank 28, Qualified for **AIME**,
Certificate of Distinction, Honour Roll (Top 2.5%)

World International Mathematics Olympiad (WIMO)

Final 2024: (Kuala Lumpur, Malaysia)

Bronze Award

CEMC (University of Waterloo):

Pascal Contest (9th Grade, 2023):

2nd Place from India region, Certificate of
Distinction, Student Honour Roll

Fryer Contest (9th Grade, 2023): 2nd Place from India
region, Certificate of Distinction

Cayley Contest (10th Grade, 2024): Certificate of
Distinction, Student Honour Roll

Galios contest (10th Grade, 2024):

Certificate of Distinction

Fermat Contest (11th Grade, 2025) :

9th Place from India region, Certificate of Distinction,
Student Honour Roll

Hypatia Contest (11th Grade, 2025):

Certificate of Distinction

SIMOC (Singapore International Mathematical Olympiad Challenge) 2024:

Maths Olympiad Contest: Silver Award

Maths Warrior Challenge: Silver Award

Mind Sports Challenge: Bronze Award

IOQM (Indian Olympiad Qualifier in Mathematics)

2024-25:

Certificate of Merit

SASMO (Singapore and Asian School Mathematics Olympiad) 2024: (Grade 10)

Gold Award

AMO (American Mathematics Olympiad):

Gold Award (2023, Grade 10)

Gold Award (2022, Grade 9)

TIMO (Thailand International Mathematical Olympiad)

Final Round 2022-23:

Gold Award (Secondary 3 Group)

**HKIMO (Hong Kong International Mathematical
Olympiad) Heat Round 2024:**

Gold Award (Senior Secondary Group)

HKIMO (Hong Kong International Mathematical Olympiad) Final Round 2022

Gold Award (Secondary 2 Group)

Stanford Math Tournament (SMT) 2023:

Honourable Mention in General Test

Australian Mathematics Competition 2024:

Certificate of Distinction (Senior Division, Grade 11)

International Maths Kangaroo Competition (2023):

Silver Medal

International Junior Honor Society (IJHS), Singapore (2023/2024):

Certificate of Memb

CHEMISTRY

Avogadro Chemistry Contest(UNIVERSITY OF WATERLOO) (11th Grade, 2025):

Certificate of Distinction

PHYSICS

Sir Issac Newton Physics Contest (UNIVERSITY OF WATERLOO) (11th Grade, 2025):

Certificate of Distinction

Future Intelligence Science Olympiad Physics, International round (2024):

Gold Medal

ROBOTICS

First Tech Challenge (FTC) 2024:

Role: CAD and Design Head of Team

Achievements: Qualified for Asian Pacific Open Championship (APOC), Sydney, Australia. Won Inspire Award (2nd Place) and Think Award (2024)

F1 in Schools 2024

Role: Design Engineer

Ranked 15th in Nationals

Regional Digital Marketing Award

Social Impact Projects

Teens of God (Regd. Local NGO) (*Aug. 2022-Dec 2023*)

Mentor/Tutor

Teaching basic Math and Science concepts to underprivileged children.

One of the lead members of the NGO's robotics branch, where we taught concepts of robotics and circuits using donated STEM kits

SKILLS

Technical Skills/Tools:

Python, 3D Modelling (CAD), Arduino, Robotics

Sports:

DIPLOMA & BLACK BELT in Shotokan Karate

Music:

Guitar (Trinity Rock & Pop Grade 5)

Languages:

English, French, Hindi, Punjabi

Membership

New York Academy of Sciences- The Junior Academy member

Selected member; engaged for a year with the Academy's STEM community. Part of the Solar Power team under Energy Infrastructure.